Band-1 Goldbach: Explicit-X Verification Probe (Collapse List 1.16)

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Band-1 Goldbach: Explicit Verification Bound X (Probe Only)

Honesty stamp. This note records the **published** exhaustive verification bound for the **strong (even) Goldbach** statement and certifies a **tiny on-box re-check** of the same method on $[4, 10^6]$. It does **not** prove the strong Goldbach conjecture. It does **not** claim a Clay-millennial solution. **GATE-04** is locked out of scope. Ternary / weak Goldbach is already a theorem (Helfgott) and is **not** the target.

Strong Goldbach remains OPEN. Finite-range verification $\neq$ conjecture.

Abstract

Published peer-reviewed X: every even integer n with $4 \le n \le 4 \cdot 10^{18}$ is a sum of two primes (Oliveira e Silva–Herzog–Pardi, *Mathematics of Computation* **83** (2014), 2033–2060; DOI 10.1090/S0025-5718-2013-02787-1).

Box method demo / re-check: every even integer in $[4, 10^6]$ admits a Goldbach partition (**499{,}999** checks; **0** failures; wall ≈ 0.23 s). This window is **not** a new world record (orders of magnitude below $4 \cdot 10^{18}$).

Problem (Band-1 collapse item)

1. Cite the current **published** explicit verification bound X_{pub} for strong Goldbach without invention.
2. Re-run a classical sieve + minimal-partition scan on a **certified small box window**.
3. Keep ternary Goldbach, GATE-04, and Clay rhetoric out of the claim surface.

Literature ledger (2024–2026 web check; not re-proved on box)

Year	Limit	Authors	Status
1855	10^5	Desboves (Haussner 1896)	historical
1938	10^5	Pipping	historical
1993	$4 \cdot 10^{11}$	Sinisalo	published
1998	10^{14}	Deshouillers, te Riele, Saouter	published (Saouter lineage)
2001	$4 \cdot 10^{14}$	Richstein, *Math. Comp.* 70	published
2014	$4 \cdot 10^{18}$	**Oliveira e Silva, Herzog, Pardi**, *Math. Comp.* 83	**published record**
2015	n/a	Platt–Trudgian (Linnik K / powers of two)	adjacent theory — **not** an X for strong Goldbach
2023–2026	claims $> 4 \cdot 10^{18}$	Zenodo / Medium Gridbach / GPU preprints	**not** elevated above Math. Comp. record

Canonical citation for $X_{\text{pub}} = 4 \cdot 10^{18}$. T. Oliveira e Silva, S. Herzog, S. Pardi, *Empirical verification of the even Goldbach conjecture and computation of prime gaps up to* $4 \cdot 10^{18}$, Mathematics of Computation **83** (2014), 2033–2060. Author page: https://sweet.ua.pt/tos/goldbach.html. Wikipedia (accessed Sep 2026 literature sweep) agrees: verified for $n \leq 4 \times 10^{18}$ (double-checked to 4×10^{17}).

Hygiene. Informal / Zenodo / Medium claims past $4 \cdot 10^{18}$ are ledger-noted and **not** treated as the published X. Platt–Trudgian work in the Linnik-approximation lane does **not** replace OHS as the verification bound.

Ternary context (not claimed). Weak / odd Goldbach is settled for large n by Helfgott; OHS + Ramaré–Saouter historically pushed odd-Goldbach numerical ranges, but that is separate from tonight's strong-Goldbach X probe.

Method (box)

1. **Sieve.** Eratosthenes boolean array `is_prime[0..X]` for $X = 10^6$.
2. **Minimal partition.** For each even $n = 4, 6, \ldots, X$, find the smallest prime $p \leq n/2$ with $n - p$ prime.
3. **Unit tests.** primes ≤ 30; $\pi(10^6) = 78498$; full verify on $[4, 30]$.
4. **Label.** Method demo / re-check — **not** a record attempt.

Results

Published X (cited)

Quantity	Value
X_{pub}	$4 \cdot 10^{18}$
Source	OHS, *Math. Comp.* 83 (2014)
Box beats published?	**No**

Box window (certified tonight)

Quantity	Value
Box X	10^6
Range	$[4, 10^6]$
Even integers checked	**499{,}999**
Failures	**0**
All verified	**True**
$\pi(10^6)$	**78{,}498**
Max minimal prime p	**523** at $n = 503{,}222$
Sieve time	≈ 0.05 s
Verify time	≈ 0.18 s
Wall (box)	≈ 0.23 s

Sample minimal partitions

n	p	$n - p$
4	2	2
10	3	7
100	3	97
1000	3	997
10000	59	9941
100000	11	99989

n	p	$n-p$
1000000	17	999983

Why Band-1 vs conjecture

- **Band-1 (this probe):** cite explicit published X; certify a finite re-check window; method hygiene.
- **Conjecture (OPEN):** every even integer ≥ 4 is a sum of two primes — **not proved here**.

Reproducibility

- Lab: `/workspace/mission-control/experiments/band1/goldbach_explicit_X/`
- Run: `python3 goldbach_explicit_X.py`
- Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `METHOD.md`

Conclusion

Published X: $4 \cdot 10^{18}$ (Oliveira e Silva–Herzog–Pardi 2014). **Box window:** every even $n \in [4, 10^6]$ is a sum of two primes (0 failures). **Strong Goldbach: OPEN**. No Clay claim. GATE-04 locked.